



10.0.1 UGC NET CSE | June 2013 | Part 2 | Question: 38



How many edges must be removed to produce the spanning forest of a graph with N vertices, M edges and C connected components?

- A. $M+N-C$ B. $M-N-C$ C. $M-N+C$ D. $M+N+C$

ugcnetcse-june2013-paper2

Answer key

10.0.2 UGC NET CSE | June 2013 | Part 2 | Question: 37



1. Which of the following connected graph has exactly one spanning tree?

- A. Complete graph B. Hamiltonian graph
C. Euler graph D. None of the above

graph-theory ugcnetcse-june2013-paper2

Answer key

10.0.3 UGC NET CSE | June 2014 | Part 3 | Question: 65



Given the following statements :

S_1 : The subgraph-isomorphism problem takes two graphs G_1 and G_2 and asks whether G_1 is a subgraph of G_2

.

S_2 : The set-partition problem takes as input a set S of numbers and asks whether the numbers can be partitioned into two sets A and $\bar{A} = S - A$ such that

$$\sum_{x \in A} x = \sum_{x \in \bar{A}} x$$

Which of the following is true ?

- A. S_1 is NP problem and S_2 is P problem.
B. S_1 is NP problem and S_2 is NP problem.
C. S_1 is P problem and S_2 is P problem.
D. S_1 is P problem and S_2 is NP problem.

ugcnetjune2014iii graph-theory

Answer key

10.0.4 UGC NET CSE | June 2013 | Part 2 | Question: 5



Cyclometric complexity of a flow graph G with n vertices and e edges is

- A. $V(G) = e+n-2$ B. $V(G) = e-n+2$ C. $V(G) = e+n+2$ D. $V(G) = e-n-2$

ugcnetcse-june2013-paper2 graph-theory

Answer key

10.0.5 UGC NET CSE | June 2013 | Part 2 | Question: 43



Which one of the following is decimal value of a signed binary number 1101010, if it is in 2's complement

form?

- A. -42 B. -22 C. -21 D. -106

ugcnetcse-june2013-paper2

Answer key 

10.0.6 UGC NET CSE | June 2013 | Part 2 | Question: 45



When a mobile telephone physically moves from one to another cell, the base station transfers ownership to the cell getting strongest signal. This process is known as

- A. handoff B. mobile switching
C. mobile routing D. cell switching

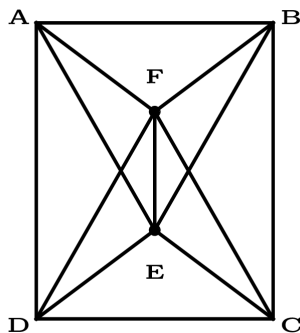
ugcnetcse-june2013-paper2

Answer key 

10.0.7 UGC NET CSE | December 2014 | Part 2 | Question: 03



Consider the Graph shown below :



This graph is a _____.

- A. Complete Graph B. Bipartite Graph
C. Hamiltonian Graph D. All of the above

ugcnetcse-dec2014-paper2 graph-theory

Answer key 

10.0.8 UGC NET CSE | December 2013 | Part 3 | Question: 39



A _____ complete subgraph and a _____ subset of vertices of a graph $G=(V,E)$ are a clique and a vertex cover respectively

- A. minimal, maximal B. minimal, minimal
C. maximal, minimal D. maximal, maximal

ugcnetcse-dec2013-paper3 engineering-mathematics discrete-mathematics graph-theory

Answer key 

10.0.9 UGC NET CSE | Junet 2015 | Part 2 | Question: 5



Consider a Hamiltonian Graph(G) with no loops and parallel edges. Which of the following is true with respect to this graph (G)?

- i. $\deg(v) \geq n/2$ for each vertex of G
ii. $|E(G)| \geq 1/2(n-1)(n-2) + 2$ edges
iii. $\deg(v) + \deg(w) \geq n$ for every v and w not connected by an edge

A. *i* and *ii*

B. *ii* and *iii*

C. *i* and *iii*

D. *i*, *ii*, and *iii*

ugcnetcse-june2015-paper2 discrete-mathematics graph-theory

Answer key 

10.0.10 UGC NET CSE | December 2015 | Part 2 | Question: 2



Which of the following statement(s) is/are false?

- i. A connected multigraph has an Euler Circuit if and only if each of its vertices has even degree.
- ii. A connected multigraph has an Euler Path but not an Euler Circuit if and only if it has exactly two vertices of odd degree.
- iii. A complete graph (K_n) has a Hamilton Circuit whenever $n \geq 3$.
- iv. A cycle over six vertices (C_6) is not a bipartite graph but a complete graph over 3 vertices is bipartite.

Codes :

A. i only

B. ii and iii

C. iii only

D. iv only

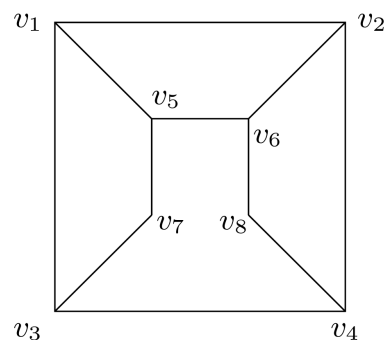
graph-theory ugcnetcse-dec2015-paper2

Answer key 

10.0.11 UGC NET CSE | December 2015 | Part 2 | Question: 4



Consider the graph given below:



The two distinct set of vertices, which make the graph bipartite are

- A. $(v_1, v_4, v_6); (v_2, v_3, v_5, v_7, v_8)$
- B. $(v_1, v_7, v_8); (v_2, v_3, v_5, v_6)$
- C. $(v_1, v_4, v_6, v_7); (v_2, v_3, v_5, v_8)$
- D. $(v_1, v_4, v_6, v_7, v_8); (v_2, v_3, v_5)$

ugcnetcse-dec2015-paper2 graph-theory

Answer key 

10.0.12 UGC NET CSE | December 2009 | Part 2 | Question: 02



Circle has _____

- (A) No vertices
- (B) Only 1 vertex
- (C) ∞ vertices
- (D) None of these

ugcnetcse-dec2009-paper2 graph-theory

Answer key 

10.0.13 UGC NET CSE | December 2009 | Part 2 | Question: 22



The number of edges in a complete graph of n vertices is

- (A) n
- (B) $n(n-1)/2$
- (C) $n(n+1)/2$
- (D) $(n^2)/2$

ugcnetcse-dec2009-paper2 graph-theory

Answer key 

10.0.14 UGC NET CSE | June 2009 | Part 2 | Question: 12



The complete graph with four vertices has k edges where k is :

- A. 3
- B. 4
- C. 5
- D. 6

ugcnetcse-june2009-paper2

Answer key 

10.0.15 UGC NET CSE | June 2009 | Part 2 | Question: 23



Which two of the following are equivalent for an undirected graph G ?

- (i) G is a tree
- (ii) There is at least one path between any two distinct vertices of G
- (iii) G contains no cycles and has $(n-1)$ edges
- (iv) G has n edges

- | | |
|-----------------|-------------------|
| A. (i) and (ii) | B. (i) and (iii) |
| C. (i) and (iv) | D. (ii) and (iii) |

ugcnetcse-june2009-paper2

Answer key 

10.0.16 UGC NET CSE | December 2008 | Part 2 | Question: 2



The graph $K_{3,4}$ has:

- | | |
|------------|-------------|
| A. 3 edges | B. 4 edges |
| C. 7 edges | D. 12 edges |

ugcnetcse-dec2008-paper2 graph-theory

Answer key 

10.0.17 UGC NET CSE | July 2018 | Part 2 | Question: 81



E is the number of edges in the graph and f is maximum flow in the graph. When the capacities are integers, the runtime of Ford-Fulberson algorithm is bounded by

- | | | | |
|---------------|-----------------|-----------------|-------------------|
| A. $O(E * f)$ | B. $O(E^2 * f)$ | C. $O(E * f^2)$ | D. $O(E^2 * f^2)$ |
|---------------|-----------------|-----------------|-------------------|

ugcnetcse-july2018-paper2 graph-theory

- A. $m \neq n, m, n \geq 2$
C. $m = n, m, n \geq 2$

ugcnetcse-june2019-paper2 graph-theory

- A. Is a bi-directional graph
B. Is directed graph
C. Is graph in which number associated with arc
D. Eliminates table method

ugcnetcse-dec2004-paper2

- 1, 2, 3, 4, 5
- 3, 4, 5, 6, 7
- 1, 4, 5, 8, 6
- 3, 4, 5, 6

- A. (i) and (ii) B. (iii) and (iv) C. (iii) and (ii) D. (ii) and (iv)

ugcnetcse-dec2004-paper2

- A. $N(N-1)$ B. $2N-1$ C. $N-1$ D. $N(N-1)/2$

ugcnetcse-dec2006-paper2

- A. $n(n-1)$ B. $\frac{n(n-1)}{2}$ C. n^2 D. $2n-1$

ugcnetcse-dec2007-paper2

Binary Tree (1)

A. $e = i + n$

B. $e = i + 2n$

C. $e = 2i + n$

D. $e = 2n + i$

ugcnetcse-nov2017-paper3 data-structures binary-tree

Answer key 

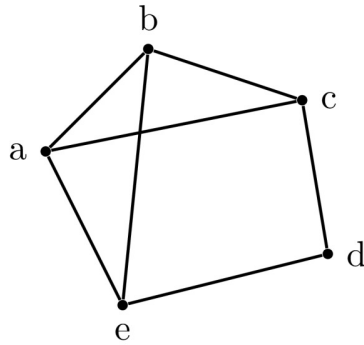
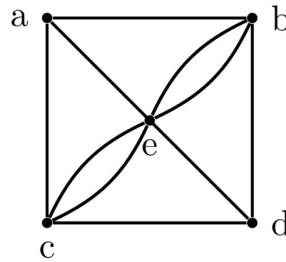
10.2

Euler Graph (2)

10.2.1 Euler Graph: UGC NET CSE | August 2016 | Part 2 | Question: 5



Given the following graphs :

 (G_1)  (G_2)

Which of the following is correct ?

- A. G_1 contains Euler circuit and (G_2) does not contain Euler circuit.
 B. (G_1) does not contain Euler circuit and (G_2) contains Euler circuit.
 C. Both (G_1) and (G_2) do not contain Euler circuit.
 D. Both (G_1) and (G_2) contain Euler circuit.

ugcnetcse-aug2016-paper2 graph-theory euler-graph

Answer key 

10.2.2 Euler Graph: UGC NET CSE | December 2010 | Part 2 | Question: 2



An undirected graph possesses an eulerian circuit if and only if it is connected and its vertices are

- A. All of even degree
 B. All of odd degree
 C. Of any degree
 D. Even in number

ugcnetcse-dec2010-paper2 graph-theory euler-graph

Answer key 

10.3

Flow Network (1)

10.3.1 Flow Network: UGC NET CSE | June 2019 | Part 2 | Question: 69



Consider the following properties with respect to a flow network $G = (V, E)$ in which a flow is a real-valued function $f : V \times V \rightarrow R$:

$$P_1: \text{For all } u, v \in V, f(u, v) = -f(v, u)$$

$$P_2: \sum_{v \in V} f(u, v) = 0 \text{ for all } u \in V$$

Which one of the following is/are correct?

- A. Only P_1
 B. Only P_2
 C. Both P_1 and P_2
 D. Neither P_1 nor P_2

ugcnetcse-june2019-paper2 flow-network

Answer key 

10.4.1 Graph Coloring: UGC NET CSE | January 2017 | Part 2 | Question: 5



Consider a Hamiltonian Graph G with no loops or parallel edges and with $|V(G)| = n \geq 3$. The which of the following is true?

- A. $\deg(v) \geq \frac{n}{2}$ for each vertex v
- B. $|E(G)| \geq \frac{1}{2}(n-1)(n-2) + 2$
- C. $\deg(v) + \deg(w) \geq n$ whenever v and w are not connected by an edge
- D. All of the above

ugcnetjan2017ii graph-theory graph-coloring

Answer key

10.4.2 Graph Coloring: UGC NET CSE | June 2010 | Part 2 | Question: 4



S_1 : I teach algorithms and maths.

S_2 : My professor teaches maths, electronics and computer science.

S_3 : I have a student of maths.

S_4 : Algorithm is a part of computer science.

S_5 : Maths students know computer science.

What would be the chromatic number of a graph, vertices of which are the actors/entities that are involved in the sentences S_1 to S_5 and edges to represent the associations/relationships amongst the entities/actors as expressed in the sentences S_1 to S_5 above ?

- A. 2
- B. 3
- C. 4
- D. None of these

ugcnetcse-june2010-paper2 graph-coloring

Answer key

10.4.3 Graph Coloring: UGC NET CSE | June 2012 | Part 2 | Question: 4



The number of colours required to properly colour the vertices of every planer graph is

- A. 2
- B. 3
- C. 4
- D. 5

ugcnetcse-june2012-paper2 graph-theory graph-coloring

Answer key

10.5.1 Graph Connectivity: UGC NET CSE | August 2016 | Part 2 | Question: 22



Consider an undirected graph G where self-loops are not allowed. The vertex set of G is $(i, j) | 1 \leq i \leq 12, 1 \leq j \leq 12$. There is an edge between (a, b) and (c, d) if $|a-c| \leq 1$ or $|b-d| \leq 1$. The number of edges in this graph is

- A. 726
- B. 796
- C. 506
- D. 616

ugcnetcse-aug2016-paper2 graph-theory graph-connectivity

Answer key

10.5.2 Graph Connectivity: UGC NET CSE | December 2010 | Part 2 | Question: 4



The minimum number of edges in a connected graph with ' n ' vertices is equal to

- A. $n(n-1)$ B. $\frac{n(n-1)}{2}$ C. n^2 D. $n-1$

ugcnetcse-dec2010-paper2 graph-theory graph-connectivity

Answer key

10.5.3 Graph Connectivity: UGC NET CSE | December 2011 | Part 2 | Question: 15



McCabe's cyclomatic metric $V(G)$ of a graph G with n vertices, e edges and p connected component is

- A. e B. n C. $e-n+p$ D. $e-n+2p$

ugcnetcse-dec2011-paper2 graph-theory graph-connectivity

Answer key

10.5.4 Graph Connectivity: UGC NET CSE | December 2011 | Part 2 | Question: 5



Maximum number of edges in a n -Node undirected graph without self loop is

- A. n^2 B. $n(n-1)$
C. $n(n+1)$ D. $\frac{n(n-1)}{2}$

ugcnetcse-dec2011-paper2 graph-theory graph-connectivity

Answer key

10.5.5 Graph Connectivity: UGC NET CSE | June 2013 | Part 3 | Question: 15



A vertex cover of an undirected graph $G(V, E)$ is a subset $V_1 \subseteq V$ vertices such that

- A. Each pair of vertices in V_1 is connected by an edge
B. If $(u, v) \in E$ then $u \in V_1$ and $v \in V_1$
C. If $(u, v) \in E$ then $u \in V_1$ or $v \in V_1$
D. All pair of vertices in V_1 are not connected by an edge

ugcnetcse-june2013-paper3 graph-theory graph-connectivity

Answer key

10.5.6 Graph Connectivity: UGC NET CSE | September 2013 | Part 2 | Question: 22



A simple graph G with n vertices is connected if the graph has

- A. $(n-1)(n-2)/2$ edges B. more than $(n-1)(n-2)/2$ edges
C. less than $(n-1)(n-2)/2$ edges D. $\sum_{i=1}^k C(n_i, 2)$ edges

ugcnetsep2013iii graph-theory graph-connectivity

Answer key

10.5.7 Graph Connectivity: UGC NET CSE | September 2013 | Part 2 | Question: 8



Consider an undirected graph G with 100 nodes. The maximum number of edges to be included in G so that the graph is not connected is

- A. 2451 B. 4950 C. 4851 D. 9900

ugcnetsep2013iii graph-theory graph-connectivity

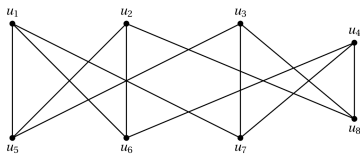
Answer key

10.6

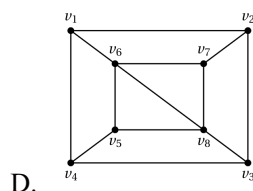
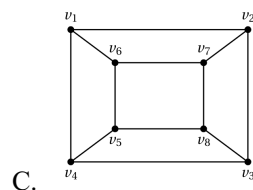
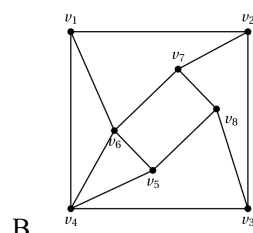
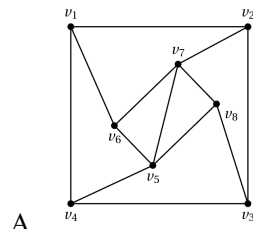
Graph Isomorphism (1)



Consider the graph given below as :



Which one of the following graph is isomorphic to the above graph ?



ugcnetcse-june2014-paper2 graph-theory non-gatecse graph-isomorphism

Answer key

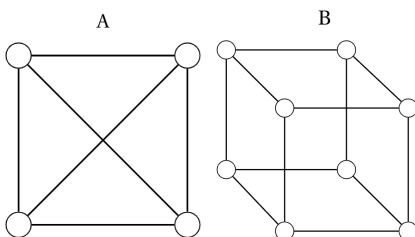
10.7

Graph Planarity (2)

10.7.1 Graph Planarity: UGC NET CSE | December 2012 | Part 3 | Question: 46



Two graphs A and B are shown below: Which one of the following statements is true?



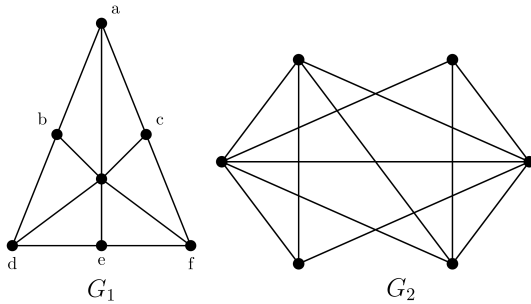
- A. Both A and B are planar
C. A is planar and B is not

- B. Neither A nor B is planar
D. B is planar and A is not

ugcnetcse-dec2012-paper3 graph-theory graph-planarity

Answer key 

10.7.2 Graph Planarity: UGC NET CSE | June 2012 | Part 3 | Question: 72



G_1 and G_2 are two graphs as shown:

- A. Both G_1 and G_2 are planar graphs
B. Both G_1 and G_2 are not planar graphs
C. G_1 is planar and G_2 is not planar
D. G_1 is not planar and G_2 is planar

ugcnetcse-june2012-paper3 graph-theory graph-planarity

Answer key 

10.8 Hamiltonian Graph (1)

10.8.1 Hamiltonian Graph: UGC NET CSE | June 2014 | Part 2 | Question: 23



Consider a complete bipartite graph $K_{m,n}$. For which values of m and n does this, complete graph have a Hamilton circuit

- A. $m = 3, n = 2$
B. $m = 2, n = 3$
C. $m = n > 2$
D. $m = n > 3$

ugcnetcse-june2014-paper2 graph-theory hamiltonian-graph

Answer key 

10.9 Handshaking Theorem (1)

10.9.1 Handshaking Theorem: UGC NET CSE | June 2019 | Part 2 | Question: 4



Suppose that a connected planar graph has six vertices, each of degree four. Into how many regions is the plane divided by a planar representation of this graph?

- A. 6
B. 8
C. 12
D. 20

ugcnetcse-june2019-paper2 graph-planarity handshaking-theorem

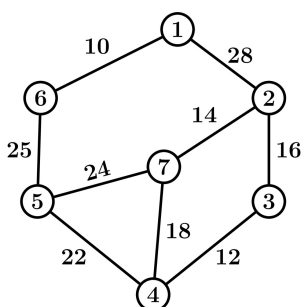
Answer key 

10.10 Minimum Spanning Tree (1)

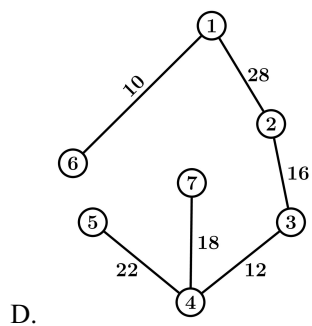
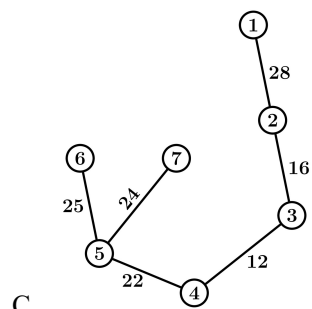
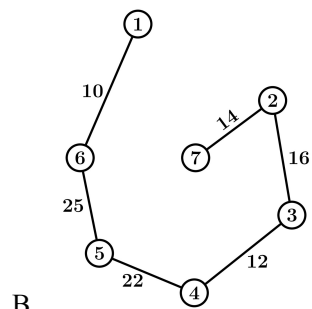
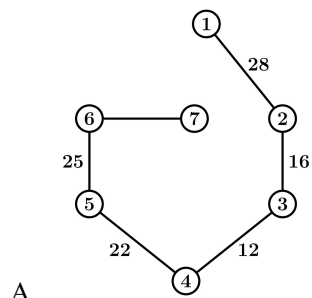
10.10.1 Minimum Spanning Tree: UGC NET CSE | Junet 2015 | Part 2 | Question: 21



Consider the given graph



Its Minimum Cost Spanning Tree is _____



ugcnetcse-june2015-paper2 graph-theory minimum-spanning-tree

Answer key

10.11

Non Planar (1)

10.11.1 Non Planar: UGC NET CSE | December 2013 | Part 2 | Question: 39



A graph is non-planar if and only if it contains a subgraph homomorphic to

- A. $K_{3,2}$ or K_5
- B. $K_{3,3}$ or K_6
- C. $K_{3,3}$ or K_5
- D. $K_{2,3}$ or K_5

ugcnetcse-dec2013-paper2 graph-theory non-planar

Answer key

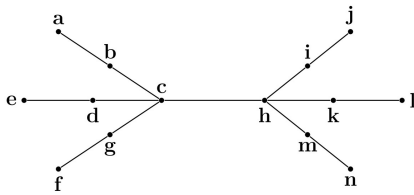
10.12

Tree (2)

10.12.1 Tree: UGC NET CSE | December 2012 | Part 2 | Question: 33



Consider the tree given below:



Using the property of eccentricity of a vertex, find every vertex that is the centre of the given tree:

- A. d & h B. c & k C. g, b, c, h, i, m D. c & h

ugcnetcse-dec2012-paper2 graph-theory tree graph-connectivity

Answer key

10.12.2 Tree: UGC NET CSE | December 2013 | Part 2 | Question: 36



How many edges are there in a forest of t-trees containing a total of n vertices ?

- A. $n + t$ B. $n - t$ C. $n * t$ D. n^t

ugcnetcse-dec2013-paper2 data-structures tree

Answer key

Answer Keys

10.0.1	C	10.0.2	D	10.0.3	B	10.0.4	B	10.0.5	B
10.0.6	A	10.0.7	C	10.0.8	C	10.0.9	D	10.0.10	D
10.0.11	C	10.0.12	Q-Q	10.0.13	Q-Q	10.0.14	Q-Q	10.0.15	Q-Q
10.0.16	Q-Q	10.0.17	Q-Q	10.0.18	C	10.0.19	Q-Q	10.0.20	Q-Q
10.0.21	Q-Q	10.0.22	Q-Q	10.1.1	B	10.2.1	C	10.2.2	A
10.3.1	A	10.4.1	D	10.4.2	Q-Q	10.4.3	C	10.5.1	Q-Q
10.5.2	Q-Q	10.5.3	D	10.5.4	D	10.5.5	C	10.5.6	B
10.5.7	C	10.6.1	C	10.7.1	A	10.7.2	D	10.8.1	C
10.9.1	B	10.10.1	B	10.11.1	C	10.12.1	D	10.12.2	B